

Line Follower project

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LINE FOLLOWING ROBOT USING ARDUINO AND L298N MOTOR DRIVER

1. Introduction

Automation and robotics play an important role in modern engineering applications such as industrial automation, transportation systems, and intelligent machines. One of the fundamental applications in robotics is a **line-following robot**, which is capable of following a predefined path using sensors and control logic.

This project focuses on designing and implementing a **line-following robot using an Arduino Uno, L298N motor driver, and infrared (IR) sensors**. The robot detects a black line on a white surface and adjusts its movement accordingly using real-time sensor feedback.

2. Objectives

The main objectives of this project are:

1. To design and implement a line-following robot using Arduino.
2. To understand how IR sensors detect surface colors.
3. To learn motor direction and speed control using an H-bridge motor driver.
4. To implement basic decision-making logic using sensor inputs.
5. To integrate hardware and software into a functional embedded system.

3. Components Used

Component	Quantity
Arduino Uno	1
L298N Motor Driver Module	1
DC Motors	2
IR Sensors	2
12V Battery Pack	1
Jumper Wires	As required
Robot Chassis	1

4. System Overview

The system consists of four main blocks:

1. **Power Supply Unit** – Supplies power to the motors and electronics
2. **Input Unit (IR Sensors)** – Detects the black line
3. **Control Unit (Arduino Uno)** – Processes sensor data and makes decisions
4. **Output Unit (Motor Driver & Motors)** – Executes movement

5. Power Supply Configuration

- Battery positive terminal → 12V input of L298N
- Battery negative terminal → GND of L298N
- Arduino GND → L298N GND (common ground)
- Arduino 5V → VCC of both IR sensors

Common grounding is necessary to maintain a reference voltage between all components.

6. Motor Driver (L298N) Connections

Right Motor Connections

- OUT1 → Right motor terminal 1
- OUT2 → Right motor terminal 2
- IN1 → Arduino pin 7
- IN2 → Arduino pin 8
- ENA → Arduino pin 6 (PWM)

Left Motor Connections

- OUT3 → Left motor terminal 1
- OUT4 → Left motor terminal 2
- IN3 → Arduino pin 9
- IN4 → Arduino pin 10
- ENB → Arduino pin 5 (PWM)

The enable pins (ENA and ENB) allow speed control using PWM signals.

7. IR Sensor Connections

Sensor	Arduino Pin
Right IR Sensor	Pin 11
Left IR Sensor	Pin 12

Sensor Logic:

- **Black surface** → **HIGH**
- **White surface** → **LOW**

8. Working Principle

The robot continuously reads sensor values and adjusts its movement accordingly:

Left Sensor	Right Sensor	Robot Action
LOW	LOW	Move forward
HIGH	LOW	Turn left
LOW	HIGH	Turn right
HIGH	HIGH	Stop

This allows the robot to follow the black path accurately.

9. Motor Control Logic

Motor direction is controlled using the L298N H-bridge configuration:

IN1	IN2	Motor Action
HIGH	LOW	Forward
LOW	HIGH	Reverse
LOW	LOW	Stop

Motor speed is controlled using Pulse Width Modulation (PWM) on the enable pins.

10. Software Description

10.1 Pin Definitions

The program defines pins for IR sensors, motor control, and speed control.

10.2 PWM Configuration

A register-level configuration is used to increase PWM frequency for smoother motor operation.

```
TCCR0B = TCCR0B & B11111000 | B00000010;
```

This line modifies the timer prescaler, improving motor response and reducing noise.

10.3 Main Control Logic

The program continuously:

1. Reads IR sensor values.
2. Determines movement direction.
3. Controls motor direction and speed accordingly.

11. Algorithm (Step-by-Step)

1. Start program
2. Initialize all input and output pins
3. Read values from IR sensors
4. Compare sensor values
5. Decide movement direction
6. Control motors using L298N
7. Repeat continuously

12. Results

The robot successfully follows the black line on a white surface.
It correctly adjusts direction during turns and stops when both sensors detect the black line.

13. Advantages

- Simple and cost-effective design
- Easy to understand and modify
- Suitable for beginners in robotics

14. Limitations

- Not suitable for high-speed tracking
- Sensitive to lighting conditions
- No obstacle avoidance